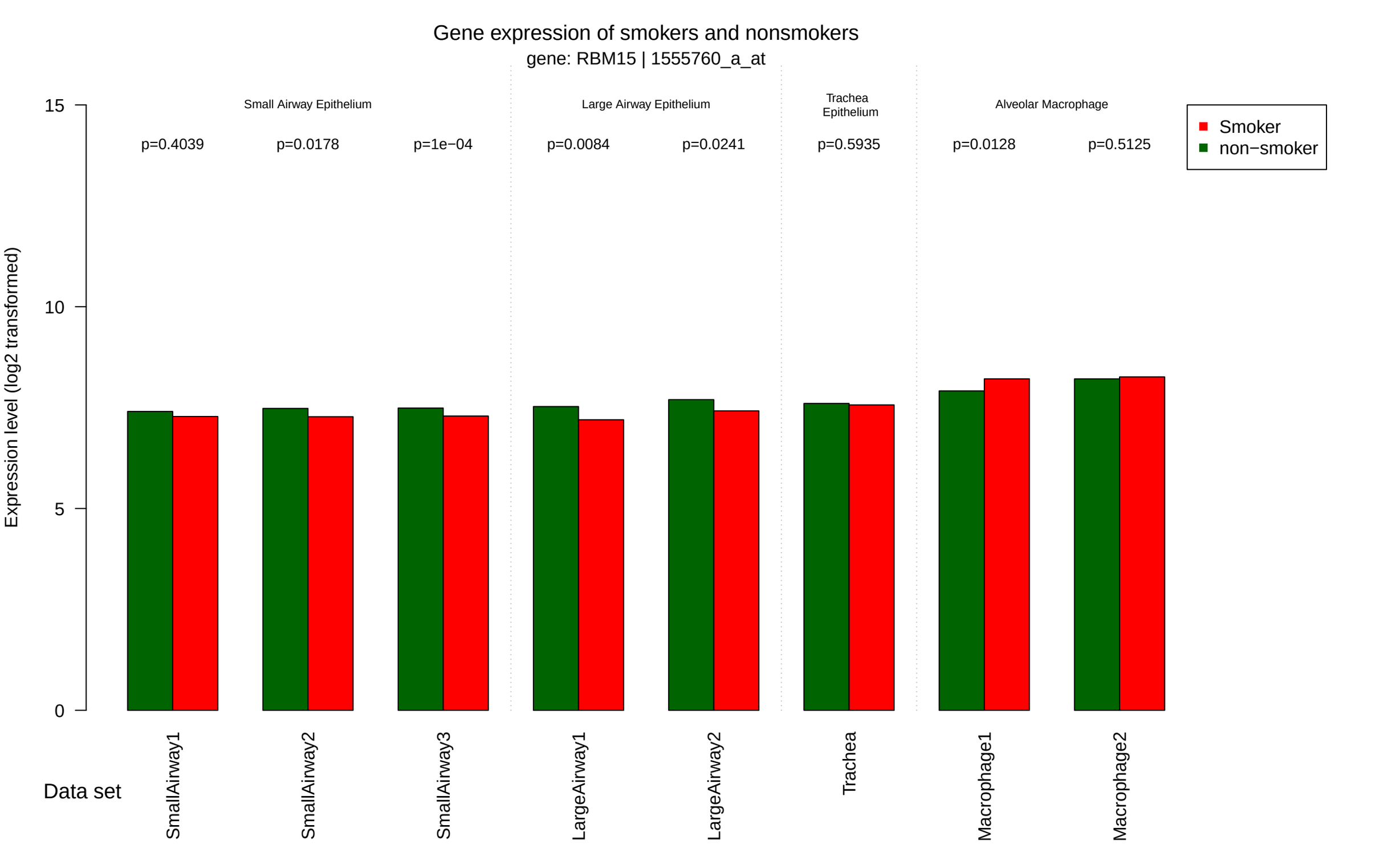
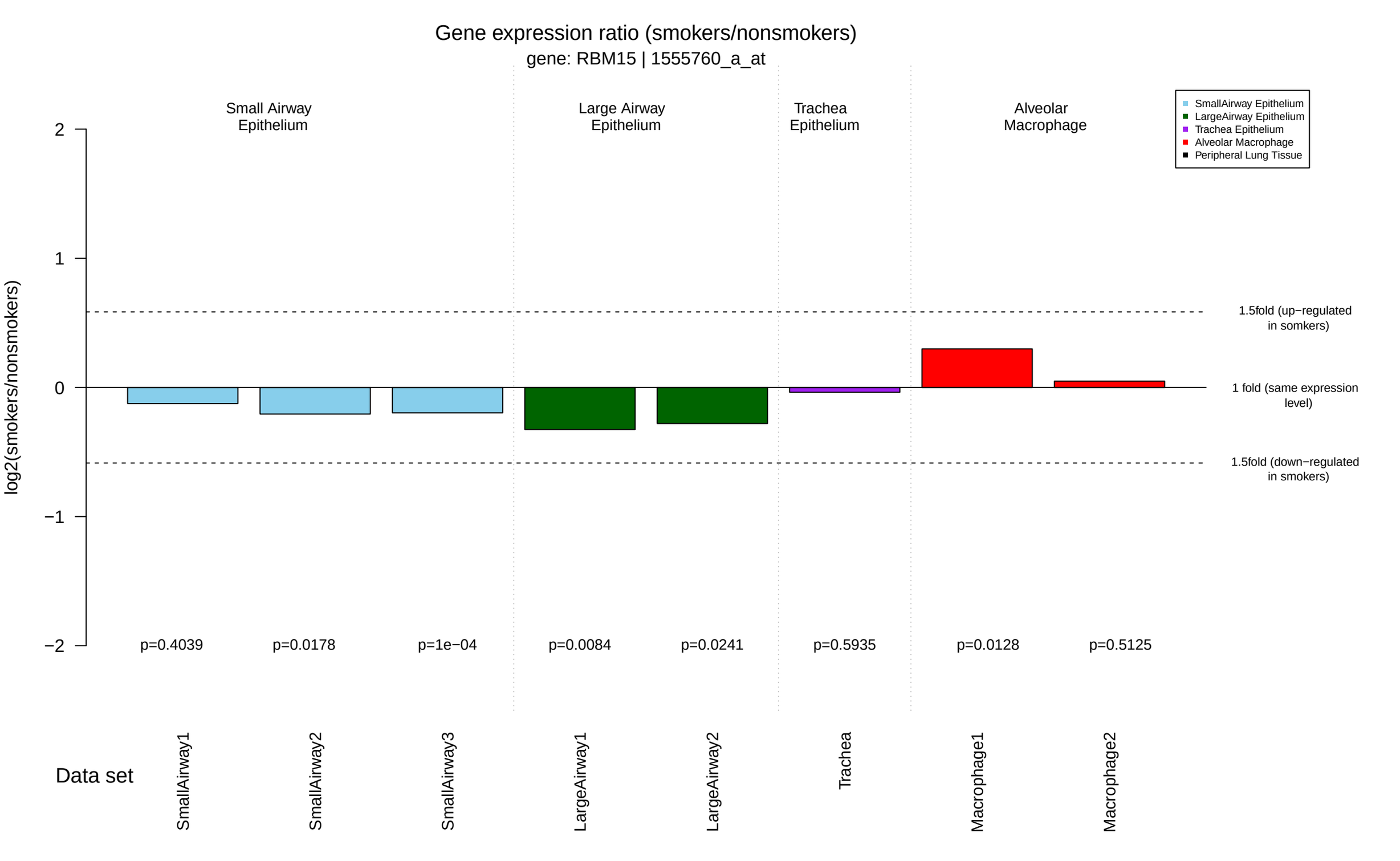
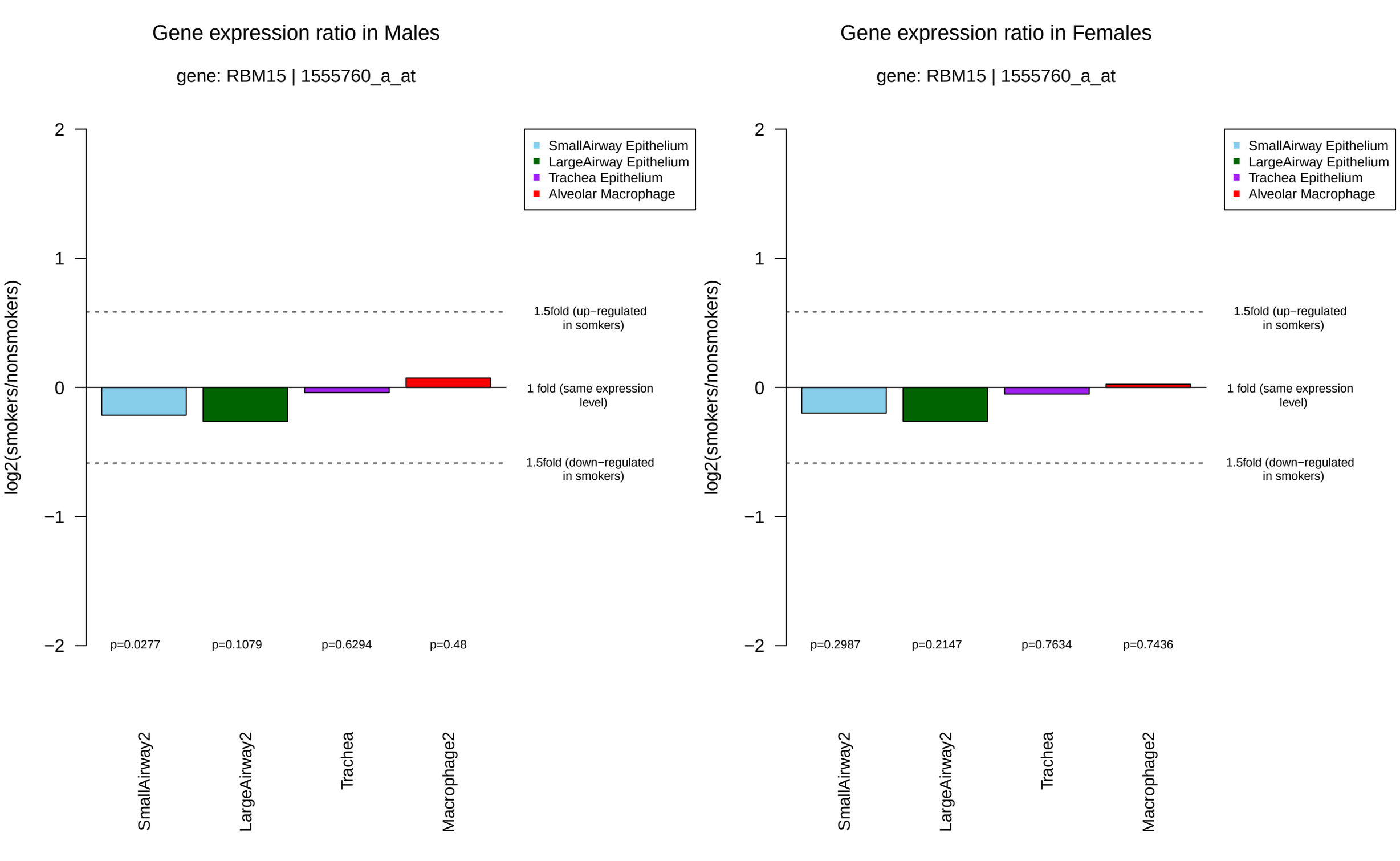
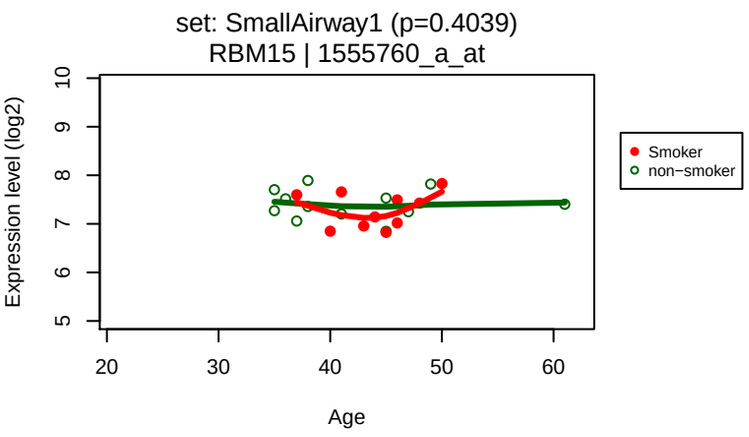
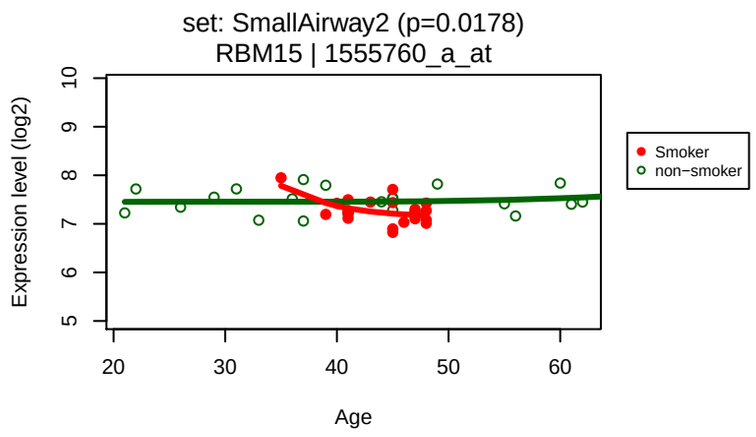
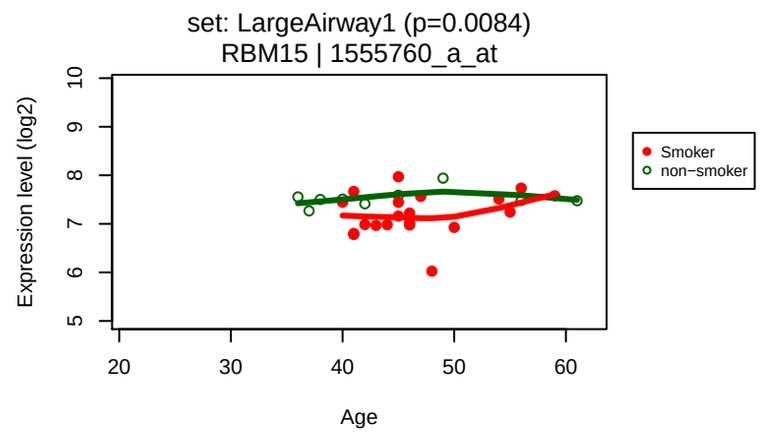

Gene expression of smokers and nonsmokers
gene: RBM15 | 1555760_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4039
p=0.0178
p=1e−04
p=0.0084
p=0.0241
p=0.5935
p=0.0128
p=0.5125
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

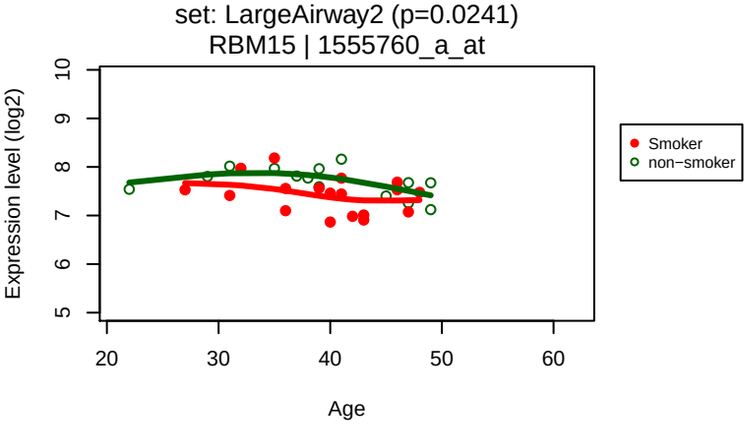

Gene expression ratio (smokers/nonsmokers)
gene: RBM15 | 1555760_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4039
p=0.0178
p=1e−04
p=0.0084
p=0.0241
p=0.5935
p=0.0128
p=0.5125
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

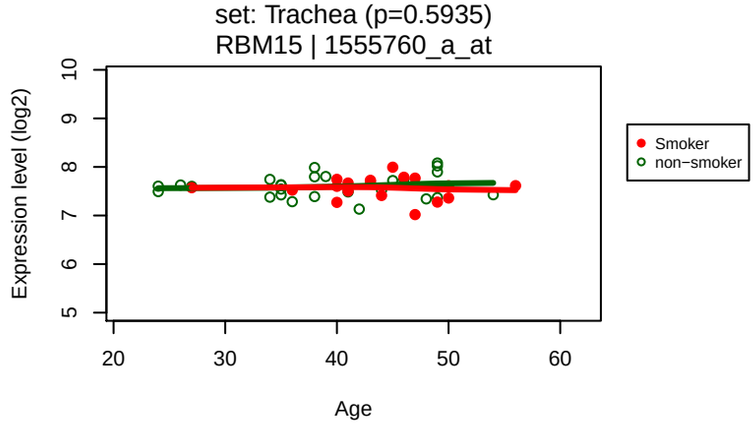

Gene expression ratio in Males
gene: RBM15 | 1555760_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0277
p=0.1079
p=0.6294
p=0.48
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM15 | 1555760_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2987
p=0.2147
p=0.7634
p=0.7436
SmallAirway2
LargeAirway2
Trachea
Macrophage2

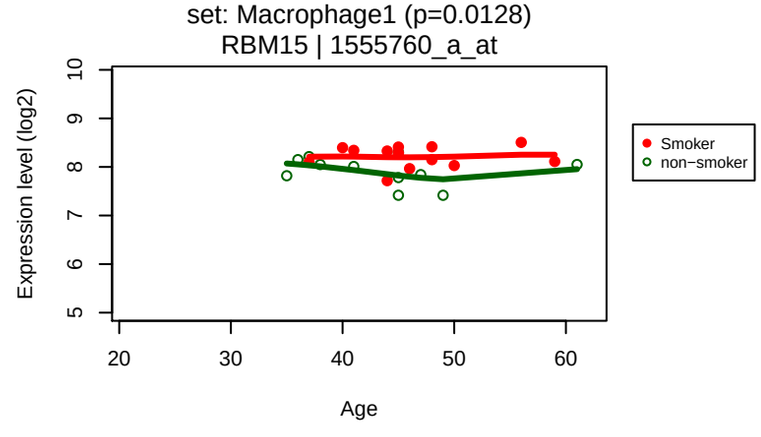

set: SmallAirway1 (p=0.4039)
RBM15 | 1555760_a_at
Expression level (log2)
Age
Smoker
non−smoker

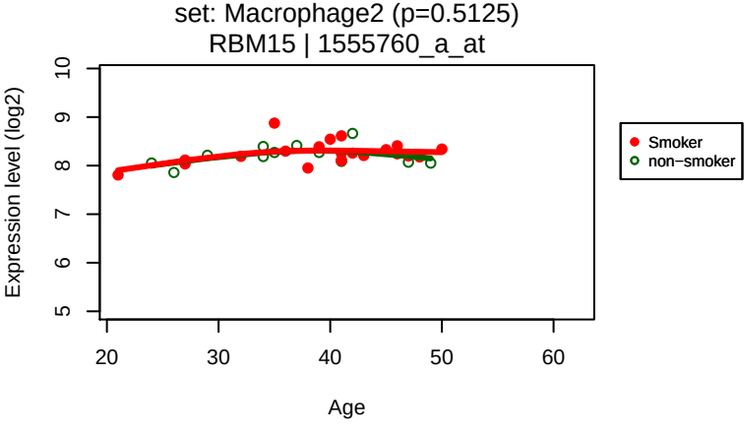

set: SmallAirway2 (p=0.0178)
RBM15 | 1555760_a_at
Expression level (log2)
Age
Smoker
non−smoker

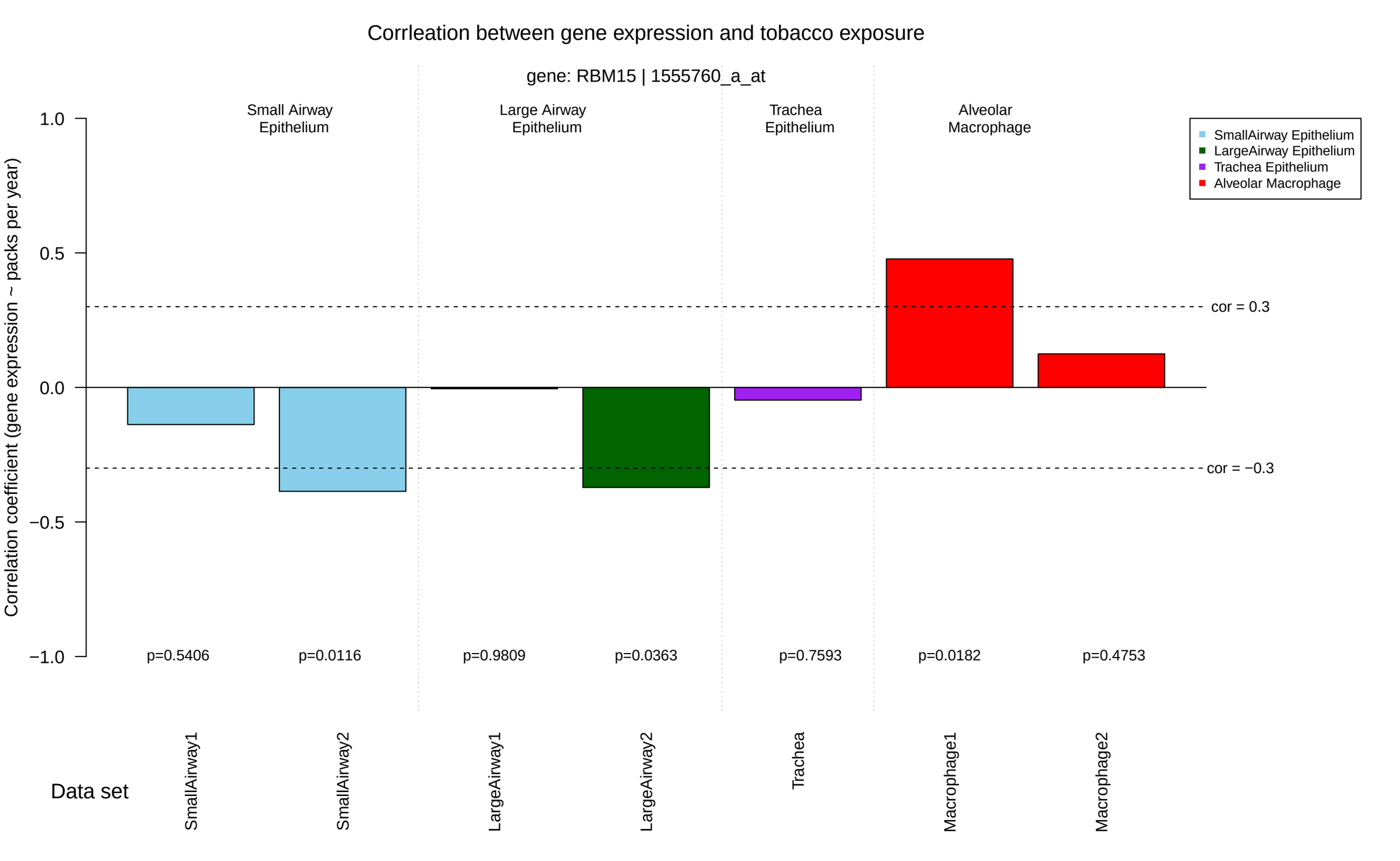

set: LargeAirway1 (p=0.0084)
RBM15 | 1555760_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0241)
RBM15 | 1555760_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5935)
RBM15 | 1555760_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0128)
RBM15 | 1555760_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5125)
RBM15 | 1555760_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM15 | 1555760_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5406
p=0.0116
p=0.9809
p=0.0363
p=0.7593
p=0.0182
p=0.4753
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2